

If $f(z) = \sinh(z)$
 verify that the functions $f(z)$ is entire
 كلية

sd $z = x + iy$

$$f(z) = \sinh(z)$$

$$= \sinh(x + iy)$$

$$\sinh(iy) = i \sinh(y)$$

$$= \sinh(x) \cosh(iy) + \sinh(iy) \cosh(x)$$

$$\therefore \cosh(iy) = \cos(y)$$

$$f(z) = \sinh(x) \cos(y) + i \sin(y) \cosh(x)$$

$$u = \sinh(x) \cos(y)$$

$$v = \sin y \cosh(x)$$

$$\frac{\partial u}{\partial x} = \cosh(x) \cos(y)$$

$$\frac{\partial v}{\partial x} = \sin y \sinh(x)$$

$$\frac{\partial u}{\partial y} = -\sinh(x) \sin(y)$$

$$\frac{\partial v}{\partial y} = \cos(y) \cosh(x)$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial y}$$

$f(z)$ analytic
 تحليلية

$f(z)$ entire

continuous

everywhere
 في كل مكان

$f(z) = z^2$	$f(z) = \bar{z}$
$f(z) = \sin(z)$	$f(z) = z ^2$
$f(z) = e^z$	$f(z) = \frac{z^2}{2}$

تحليلية

غير تحليلية

متصلة

1A $f(z) = \cosh(z)$

verify that the function $f(z)$ is entire
كلية

sol

$$z = x + iy$$

$$f(z) = \cosh(z)$$

$$= \cosh(x + iy)$$

$$= \cosh(x) \cosh(iy) + \sinh(x) \sinh(iy)$$

$$= \cosh(x) \cos(y) + i \sinh(x) \sin(y)$$

$$\begin{aligned} \sinh(ix) \\ = i \sinh(x) \end{aligned}$$

prove that

$$u(x,y) = x^2 - y^2$$

$$v(x,y) = \frac{y}{x^2 + y^2}$$

are harmonic

but are not harmonic conjugates

sd

$$u = x^2 - y^2$$

$$\frac{\partial u}{\partial x} = 2x - 0$$

$$\frac{\partial u}{\partial x^2} = 2$$

$$\frac{\partial u}{\partial y} = 0 - 2y$$

$$\frac{\partial^2 u}{\partial y^2} = -2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$2 + (-2) = 0$$

u harmonic

$$v = \frac{y}{x^2 + y^2}$$

$$\frac{\partial v}{\partial x} = \frac{(0)(x^2 + y^2) - (2x)(y)}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 v}{\partial x^2} = \frac{(-2y)(x^2 + y^2) - (2(x^2 + y^2)(2x))(-2xy)}{(x^2 + y^2)^4}$$

$$= \frac{-2y(x^2 + y^2) [x^2 + y^2 - 4x^2]}{(x^2 + y^2)^4}$$

$$= \frac{-2y [x^2 + y^2 - 4x^2]}{(x^2 + y^2)^3} = \frac{-2y (-3x^2 + y^2)}{(x^2 + y^2)^3}$$

$$= \frac{6yx^2 - 2y^3}{(x^2 + y^2)^3}$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{6yx^2 - 2y^3}{(x^2 + y^2)^3} + \frac{-6xy^2 + 2y^3}{(x^2 + y^2)^3} = \frac{0 + 0}{(x^2 + y^2)^3} = 0 \quad v \text{ harmonic}$$

$$\frac{\partial v}{\partial y} = \frac{(1)(x^2 + y^2) - (2y)(y)}{(x^2 + y^2)^2} = \frac{x^2 + y^2 - 2y^2}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 v}{\partial y^2} = \frac{(-2y)(x^2 + y^2) - (2(x^2 + y^2)(2y))(x^2 - y^2)}{(x^2 + y^2)^4}$$

$$= \frac{(-2y)(x^2 + y^2) [x^2 + y^2 + 2(x^2 - y^2)]}{(x^2 + y^2)^4}$$

$$= \frac{-2y [x^2 + y^2 + 2x^2 - 2y^2]}{(x^2 + y^2)^3}$$

$$= \frac{-2y [3x^2 - y^2]}{(x^2 + y^2)^3} = \frac{-6x^2y + 2y^3}{(x^2 + y^2)^3}$$

prove that

$$u(x,y) = x^2 - y^2$$

$$v(x,y) = \frac{y}{x^2 + y^2}$$

are harmonic

but are not harmonic conjugates

s.d

$$\therefore u + iv = x^2 - y^2 + \left(\frac{y}{x^2 + y^2} \right) i$$

$$\therefore \frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial v}{\partial x} = \frac{-2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

$$\therefore \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

not analytic

$\therefore u, v$ not harmonic conjugates

$$v = \frac{y}{x^2 + y^2}$$

$$\frac{\partial v}{\partial x} = \frac{(0)(x^2 + y^2) - (2x)(y)}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 v}{\partial x^2} = \frac{(-2y)(x^2 + y^2)^2 - (2(x^2 + y^2)(2x))(-2xy)}{(x^2 + y^2)^4}$$

$$= \frac{-2y(x^2 + y^2) [x^2 + y^2 - 2(2x)(x)]}{(x^2 + y^2)^4}$$

$$= \frac{-2y [x^2 + y^2 - 4x^2]}{(x^2 + y^2)^3} = \frac{-2y (-3x^2 + y^2)}{(x^2 + y^2)^3}$$

$$= \frac{6yx^2 - 2y^3}{(x^2 + y^2)^3}$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{6yx^2 - 2y^3}{(x^2 + y^2)^3} + \frac{-6xy^2 + 2y^3}{(x^2 + y^2)^3} = \frac{0 + 0}{(x^2 + y^2)^3} = 0$$

$$\frac{\partial v}{\partial y} = \frac{(1)(x^2 + y^2) - (2y)(y)}{(x^2 + y^2)^2} = \frac{x^2 + y^2 - 2y^2}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 v}{\partial y^2} = \frac{(-2y)(x^2 + y^2)^2 - (2(x^2 + y^2)(2y))(x^2 - y^2)}{(x^2 + y^2)^4}$$

$$= \frac{(-2y)(x^2 + y^2) [x^2 + y^2 + 2(x^2 - y^2)]}{(x^2 + y^2)^4}$$

$$= \frac{-2y [x^2 + y^2 + 2x^2 - 2y^2]}{(x^2 + y^2)^3}$$

$$= \frac{-2y [3x^2 - y^2]}{(x^2 + y^2)^3} = \frac{-6x^2y + 2y^3}{(x^2 + y^2)^3}$$

$$\int_0^{\frac{\pi}{2}} \frac{it}{e} dt$$

$$= \frac{it}{e} \Big|_0^{\frac{\pi}{2}}$$

$$\frac{1}{i} \left(e^{i\frac{\pi}{2}} - e^0 \right)$$

$$= \frac{1}{i} \left[e^{i\frac{\pi}{2}} - (1) \right]$$

$$= \frac{1}{i} \left[\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right) - 1 \right]$$

$$= \frac{1}{i} \left[0 + i - 1 \right] = \frac{i}{i} - \frac{1}{i} = 1 - (-i) = 1+i$$

$$\int_0^1 (2-i)^2 (y) (2+i) dy$$

$$(2-i)(2-i) = 4 - 4i + i^2 = 4 - 4i - 1 = 3 - 4i$$

$$z = 2 + iy$$

$$dz = i dy$$

$$\bar{z} = 2 - iy$$

$$\frac{1}{i} = -i$$

Evaluate $\int_C (\bar{z})^2 dz$

a long

C

$$y = \frac{x}{2}$$

$$z = x + iy$$

$$z = x + i\frac{x}{2} = \left(1 + \frac{i}{2}\right)x$$

$$dz = \left(1 + \frac{i}{2}\right) dx$$

$$\bar{z} = x - i\frac{x}{2} = \left(1 - \frac{i}{2}\right)x$$

$$(\bar{z})^2 = \left(1 - \frac{i}{2}\right)^2 x^2$$

$$\int_0^2 \left(1 - \frac{i}{2}\right)^2 x^2 \left(1 + \frac{i}{2}\right) dx$$

$$z \cdot \bar{z} = x^2 + y^2$$

$$= x^2 + \left(\frac{x}{2}\right)^2$$

$$\left(1 - \frac{i}{2}\right) \left(1 - \frac{i}{2}\right) \left(1 + \frac{i}{2}\right) \int_0^2 x^2 dx$$

$$= \left(1 - \frac{i}{2}\right) \left(1 + \frac{1}{4}\right) \frac{x^3}{3} \Big|_0^2$$

$$\frac{5}{4} \left(1 - \frac{i}{2}\right) \left[\frac{8}{3} - 0\right] = \frac{10}{3} \left(1 - \frac{i}{2}\right)$$

$$\frac{10}{3} + \frac{5}{3}i$$

$$\frac{10}{3} - \frac{5}{3}i$$

$$\int_0^{\frac{\pi}{2}} \frac{it}{e} dt$$

$$= \frac{it}{e} \Big|_0^{\frac{\pi}{2}}$$

$$\frac{1}{i} \left(\frac{it}{e} \Big|_0^{\frac{\pi}{2}} \right)$$

$$= \frac{1}{i} \left[e^{i\frac{\pi}{2}} - e^0 \right]$$

$$= \frac{1}{i} \left[\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right) - 1 \right]$$

$$= \frac{1}{i} \left[i - 1 \right] = \frac{i}{i} - \frac{1}{i} = 1 - (-i) = 1 + i$$

$$\int_0^1 (2-i)^2 y (2+i) dy$$

$$(2-i)(2-i) = 4 - 4i + i^2 = 4 - 4i - 1 = 3 - 4i$$

$$(3-4i) \int_0^1 y (2+i) dy$$

$$= (3-4i) \left[\frac{1}{2} (2+i) y^2 \right]_0^1$$

$$= (3-4i) \left[\frac{1}{2} (2+i) \right]$$

$$= \frac{1}{2} (3-4i)(2+i)$$

$$= \frac{1}{2} (6 + 3i - 8i - 4) = \frac{1}{2} (-2 - 5i) = -1 - \frac{5}{2}i$$

$$\frac{1}{i} = -i$$

Evaluate $\int_0^{2+i} \left(\frac{z}{2}\right)^2 dz$

$$(0, 0) \quad (2, 1)$$

$$\frac{y-0}{x-0} = \frac{1-0}{2-0}$$

$$\frac{y}{x} = \frac{1}{2}$$

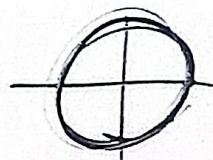
$$y = \frac{x}{2}$$

$$x = 2y$$

Evaluate

$$\int_C (z-a)^n dz$$

C circle $|z|=1$



$$(x+iy)^n = (r e^{i\theta})^n$$

$$r=1$$

$$z-a = r e^{i\theta}$$

$$dz = r i e^{i\theta} d\theta$$

$$\int_0^{2\pi} (r e^{i\theta})^n \cdot r i e^{i\theta} d\theta$$

$$(r=1)$$

$$\int_0^{2\pi} i e^{ni\theta} \cdot e^{i\theta} d\theta$$

$$i \int_0^{2\pi} e^{(n+1)i\theta} d\theta = i \int_0^{2\pi} e^{(n+1)i\theta} d\theta$$

$$\left[\frac{e^{(n+1)i\theta}}{(n+1)i} \right]_0^{2\pi}$$

$$= \frac{1}{n+1} \left[\frac{e^{(n+1)i\theta}}{e} \right]_0^{2\pi}$$

$$= \frac{1}{n+1} \left[\frac{e^{(n+1)i\theta}}{e} - e \right]$$

$$\frac{1}{n+1} [1 - 1] = 0$$

$$\begin{aligned} e^{2\pi i} &= 1 \\ e^{2\pi i} &= 1 \end{aligned}$$

